PKU Algebra II Second (Group action) Homework

1. Standard problem

II.1.1. Find a product of cyclic groups that is isomorphic to $(Z_{12} \times Z_{12})/\langle (2,6) \rangle$

pf:

Define $\varphi : Z_{12} \times Z_{12} \to Z_4 \times Z_6$

$(a, b) \mapsto (a+b \bmod 4,\ b \bmod 6)$

Clearly it's a homomorphism, surjective

$\ker \varphi = \langle (2,6) \rangle$. $\Rightarrow$ $Z_{12} \times Z_{12} / \langle (2,6) \rangle \cong Z_4 \times Z_6$ $\square$.

II.1.2. Let $\varphi: G \to H$ homomorphism. $K \leq \mathrm{Im}(\varphi)$. Show that $N_G(\varphi^{-1}(K)) = \varphi^{-1}(N_H(K))$

pf: $\forall g \in N_G(\varphi^{-1}(K))$, $g \varphi^{-1}(K) g^{-1} = \varphi^{-1}(K) \Rightarrow \varphi(g) K \varphi(g)^{-1} = K \Rightarrow \varphi(g) \in N_H(K)$

$\Rightarrow g \in \varphi^{-1}(N_H(K))$

$\forall h \in \varphi^{-1}(N_H(K)) \Rightarrow \varphi(h) K \varphi(h)^{-1} = K \Rightarrow h \varphi^{-1}(K) h^{-1} = \varphi^{-1}(K) \Rightarrow h \in N_G(\varphi^{-1}(K))$

Thus $N_G(\varphi^{-1}(K)) = \varphi^{-1}(N_H(K))$ $\square$

II.1.3. There is an injective isomorphism $i: H \to G$. ; $\pi: G \to H$ s.t. $\pi \circ i = id_H$

Show that one can write G as a semidirect product $H \ltimes \ker(\pi)$ such that i is the embedding of H into first factor H, π is the projection to the first factor

pf: $H \ltimes_\phi \ker(\pi) := \{(h, g) \mid h \in H, g \in \ker(\pi)\}$. where $\phi: H \to \mathrm{Aut}(\ker \pi)$, $\phi = \mathrm{Ad} \cdot i$

$h \mapsto \mathrm{Ad}_{i(h)}$

Then we define $\psi: H \ltimes \ker(\pi) \to G$

$(h, g) \mapsto i(h) g$.

① $\psi((h_1, g_1)(h_2, g_2)) = i(h_1 h_2)\ i(h_2)^{-1} g_1 i(h_2)\ g_2 = i(h_1) g_1 i(h_2) g_2 = \psi(h_1, g_1)\ \psi(h_2, g$

② Surjective, $\forall g_0 \in G$, consider $i(\pi(g_0))\ g_1 = g_0$ we have $\pi \circ (\ i(\pi(g_0))\ g_1 = \pi(g_0)\ \pi($

$= \pi(g_0$

Thus $g_1 \in \ker(\pi)$ Namely $\psi(\pi(g_0), g_1) = g_0$

③ Injective. $\psi(h_1, g_1) = \psi(h_2, g_2) \Rightarrow i(h_2^{-1} h_1) = g_2 g_1^{-1} \Rightarrow h_2^{-1} h_1 = \pi(g_2 g_1^{-1}) = e_H$

$\Rightarrow h_1 = h_2,\ g_1 = g_2$

Thus $G \cong H \ltimes \ker(\pi)$ $\square$

Rmk. From this, we can now view the injective homomorphism $i: H \to G$ as

$H \longrightarrow H \ltimes \ker(\pi)$ where $\pi: G \to H$. be homomorphism. $\square$.

II.1.4 Let $A = \langle x_1 \rangle \times \cdots \times \langle x_n \rangle$ be finite & Abelian with $|x_i| = n_i$

Consider the pth power map $\varphi: A \to A \quad x \mapsto x^p$

(1) Show that φ is a homomorphism

(2) Describe $\operatorname{Im}\varphi$ and $\ker\varphi$ in terms of the given generators

(3) Prove that $\ker\varphi$ and $A/\operatorname{Im}\varphi$ are elementary p-groups, namely products of $\mathbb{Z}/p\mathbb{Z}$, and they contain the same number of copies

proof (1) Obvious

(2) $\ker\varphi \cong \prod_{p \mid n_i} \langle x_i^{\frac{n_i}{p}} \rangle \qquad \operatorname{Im}\varphi = \langle x_1^p \rangle \times \langle x_2^p \rangle \times \cdots \times \langle x_n^p \rangle$

$\cong \prod_{p \mid n_i} \mathbb{Z}_{n_i/p} \times \prod_{p \nmid n_i} \mathbb{Z}_{n_i}$

(3) $\prod_{p \mid n_i} \langle x_i^{\frac{n_i}{p}} \rangle \cong (\mathbb{Z}/p\mathbb{Z})^s \quad , s = \#\{ n_i \mid p \mid n_i \}.$

Since $\ker\varphi \cong A/\operatorname{Im}\varphi$ We get the above properties. $\square$

II.1.5. For any group G, define the dual group $\hat{G}$ to be the set of all homomorphism from G into the multiplicative group of roots of unity in $\mathbb{C}$.

And the group operation $\psi \circ \chi(g) := \psi(g) \cdot \chi(g)$ where $\cdot$ is taken in $\mathbb{C}$

(1). Show that the operation on G makes $\hat{G}$ into an Abelian group with the identity $\mathrm{Id}: G \to \mathbb{C}$, $g \mapsto 1$

(2) Show that if G & H are abelian groups, then $\widehat{G \times H} \cong \hat{G} \times \hat{H}$

(3) Compute $\hat{\mathbb{Z}}_n$ as a group.

(4) If G is finite & Abelian, then $\hat{G} \cong G$.

Pf: (1) $\forall g \in G, (\chi\psi)(g) = \chi(g)\psi(g) = \psi(g)\chi(g) = (\chi\psi)(g) \Rightarrow \hat{G}$ Abelian, $\forall \psi \in \hat{G}$, $\psi^{-1}: G \to \mathbb{C}$, $g \mapsto \psi(g^{-1})$

(2) $\varphi: \hat{G} \times \hat{H} \to \widehat{G \times H}$ where $\Phi: G \times H \to \mathbb{C}$

$(\psi, \chi) \mapsto \Phi$ $\qquad (g,h) \mapsto \psi(g)\chi(h)$

It's easy to check φ is a homomorphism

① Surjective: $\forall \Phi \in \widehat{G \times H}$, $\psi_g := \Phi(g, 1)$, $\chi = \Phi(1, h)$. Then $\psi(g)\chi(h) = \Phi(g,h)$

② Injective:

$\psi_1\chi_1 = \psi_2\chi_2 \Rightarrow \psi_1(g)\chi_1(h) = \psi_2(g)\chi_2(h)$

$\Rightarrow (\psi_1\psi_2^{-1})(g) = (\chi_2\chi_1^{-1})(h)$ Fix h. then $\psi_1\psi_2^{-1}$ is a constant map $\Rightarrow \psi_1\psi_2^{-1} = \mathrm{Id}$

$\Rightarrow \psi_1 = \psi_2$, $\chi_1 = \chi_2$ Thus $\widehat{G \times H} \cong \hat{G} \times \hat{H}$

(3). $\forall \chi \in \hat{\mathbb{Z}}_n$ we have $\chi^n(g) = 1 \quad \forall g \in \mathbb{Z}_n$. $\chi(g) = e^{i\frac{2k\pi}{n}}$

let $\chi(1) = e^{i\theta}$. then $\chi(k) = e^{ik\theta}$ where $\theta = \frac{2k\pi}{n}$ $(0 \le k \le n-1)$

Let χ_0 be the identity map. χ_1 be $\mathbb{Z}_n \to \mathbb{C}$. $k \mapsto e^{\frac{2k\pi}{n}}$

$\Rightarrow \hat{\mathbb{Z}}_n = \langle \chi_1 \rangle \cong \mathbb{Z}_n$

(4) By the theorem of finite-generated Abelian group, $G \cong \mathbb{Z}_{r_1} \times \cdots \times \mathbb{Z}_{r_s}$

$\Rightarrow \hat{G} \cong \hat{\mathbb{Z}}_{r_1} \times \cdots \times \hat{\mathbb{Z}}_{r_s} \cong \mathbb{Z}_{r_1} \times \cdots \times \mathbb{Z}_{r_s}$ $\square$

$\cong G$

I.1.6 Let I be a nonempty ~~one~~ index set and let G_i be a group for each $i \in I$

The restricted direct product or direct sum of G_i is the set of all elements $(a_i)_{i\in I} \in \prod_{i\in I} G_i$ such that $a_i = 1_i$ for all but a finite number of $i \in I$

(1) Prove that the restricted product is a normal subgroup of the direct product.

(2) Let $I = \mathbb{N}$ and let p_i be the i-th prime. Show that if $G_i = \mathbb{Z}/p_i\mathbb{Z}$, then every element of restricted direct product has finite order. The restricted product is the torsion subgroup of the direct ~~subgroup~~ product.

Pf: (1) $\forall (a_i)_{i\in I} \in \prod_{i\in I} G_i$, we have $(a_i)_{i\in I}^{-1} \prod_{i\in I}' G_i (a_i)_{i\in I} = \prod_{i\in I}' (a_i^{-1} G_i a_i)$

$= \prod_{i\in I}' G_i.$

(2) $\forall (a_i)_{i\in I} \in \prod_{i\in I}' G_i$, $\langle (a_i)_{i\in I} \rangle = \langle (a_i)_{i\in I'} \rangle$ (I' be finite) with finite order.

Suppose that $(a_i)_{i\in I} \in \prod_{i\in I} G_i$ has finite order, then $\exists N$. s.t. $\forall i \in I$ $a_i^N = 1$.

$\Rightarrow \exists N > 0$. s.t. $\forall k > N$ s.t. $a_k = 1$. $\Rightarrow$ $(a_i)_{i\in I} \in \prod_{i\in I'}' G_i$

Thus the restricted product is the torsion subgroup .. □

II.1.7. Let G, H, Z be groups with embeddings $i: Z \to G$, $j: Z \to H$ s.t. $i(Z) \subseteq C(G)$, $j(Z) \subseteq C(H$

Show that (1). $\Delta: Z \longrightarrow G \times H$, $z \mapsto (i(z), j(z)^{-1})$ defines a natural embedding, and $\mathrm{Im}(\Delta) \trianglelefteq G \times H$

Denote $G \times^Z H := G \times H / \Delta(Z)$

(2) Let $G = GL_2(\mathbb{R})$, $H = \mathbb{C}^\times$. (with multiplication). and $\mathbb{Z} := \mathbb{R}^\times$

Now consider Hermitian form $\langle -, - \rangle$ with matrix $\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$ and defined on $\mathbb{C}^{\oplus 2}$

and the similitude unitary group: $GU(2) := \{ g \in GL_2(\mathbb{C}), c \in \mathbb{R}^\times \mid \langle gx, gy \rangle = c\langle x, y \rangle \}$.

Show that $GL_2(\mathbb{R}) \times^{\mathbb{R}^\times} \mathbb{C}^\times \cong GU(2)$

Pf: (1). $\Delta(z) = (i(z), j(z)^{-1})$ $(i(z_2), j(z_2)^{-1}) = (i(z_1 z_2), j(z_1 z_2)^{-1})$

$(g, h)\Delta(z)(g^{-1}, h^{-1}) = (g\, i(z) g^{-1}, h\, j(z)^{-1} h^{-1}) = (i(z), j(z)) \Rightarrow \mathrm{Im}(\Delta) \trianglelefteq G \times H$

(2) $A \in GU(2) \Leftrightarrow AA^* = \lambda I$

II 1.8. Let $H \trianglelefteq G$. For $g \in G$ consider the conjugation on H by $\varphi_g : h \mapsto ghg^{-1}$ for $h \in H$. Show that $Ad : G \to Aut\, H$, $g \mapsto \varphi_g$ is a homomorphism. Ker $Ad = C_G(H)$ $= \{g \in G \mid gh = hg \quad \forall h \in H\}$

Pf. ① $\forall g_1, g_2 \in G \quad \varphi_{g_1} \circ \varphi_{g_2}(h) = g_1 g_2 h g_2^{-1} g_1^{-1} = \varphi_{g_1 g_2}(h) \Rightarrow Ad$ is a homomorphism

② $\text{Ker } Ad = \{g \in G \mid \varphi_g = Id\} = \{g \mid ghg^{-1} = h \quad \forall h \in H\} = C_G(H)$ □

II 1.9 Let $\varphi : K \to Aut\, H$ be homomorphism. Then TFAE:

(1) The identity map $id : H \rtimes K \longrightarrow H \times K$ is a group homomorphism

(2) φ is the trivial homomorphism from K to $Aut\, H$

(3) $K \trianglelefteq H \rtimes K$

Pf: (1) $\Rightarrow$ (2): $(h_1, k_1) \circ (h_2, k_2) = (h_1 \varphi_{k_1}(h_2)\, k_1 k_2) = (h_1 \varphi_{k_1}(h_2), k_1 k_2) \Rightarrow \varphi_{k_1}(h_2) = h_2 \quad \forall k_1 \in H$

$\Rightarrow \varphi$ is trivial (2) $\Rightarrow$ (1): φ is trivial $\Rightarrow H \rtimes K \cong H \times K$

(2) $\Rightarrow$ (3): When φ is trivial, $H \times K \cong H \rtimes K$ $\forall (h, k) \in H \times K$ $(h, k)(1, K)(h^{-1}, k^{-1})$ $= (1, k)$

$\Rightarrow K = \{1\} \times K \trianglelefteq H \rtimes K$

(3) $\Rightarrow$ (1): $(h_1, k_1)(1, k_2)(h_1^{-1}, k_1^{-1}) = (h_1 \varphi_{k_1}(1), k_1 k_2)(h_1^{-1}, k_1^{-1})$

$= (h_1 \varphi_{k_1}(1)\, \varphi_{k_1 k_2}(h_1^{-1}),\ k_1 k_2 k_1^{-1})$

$\Rightarrow h_1 \varphi_{k_1}(1)\, \varphi_{k_1 k_2}(h_1^{-1}) = 1 \Rightarrow h_1 \varphi_{k_1}(1)(\varphi_{k_1} \circ \varphi_{k_2}(h_1^{-1})) = h_1 \varphi_{k_1}(\varphi_{k_2}(h_1^{-1})) = 1 \quad \forall h_1, k_1, k_2 \Rightarrow \varphi_{k_1 k_2}$ is $Id \Rightarrow \varphi$ is trivial □

II.1.10. (1) Show that under any automorphism of D_8, r has at most 2 images and r has at most 4 images. Deduce that $\#Aut(D_8) \le 8$

(2) Use the fact that $D_8 \trianglelefteq D_{16}$ to prove that $Aut(D_8) \cong D_8$

Pf: